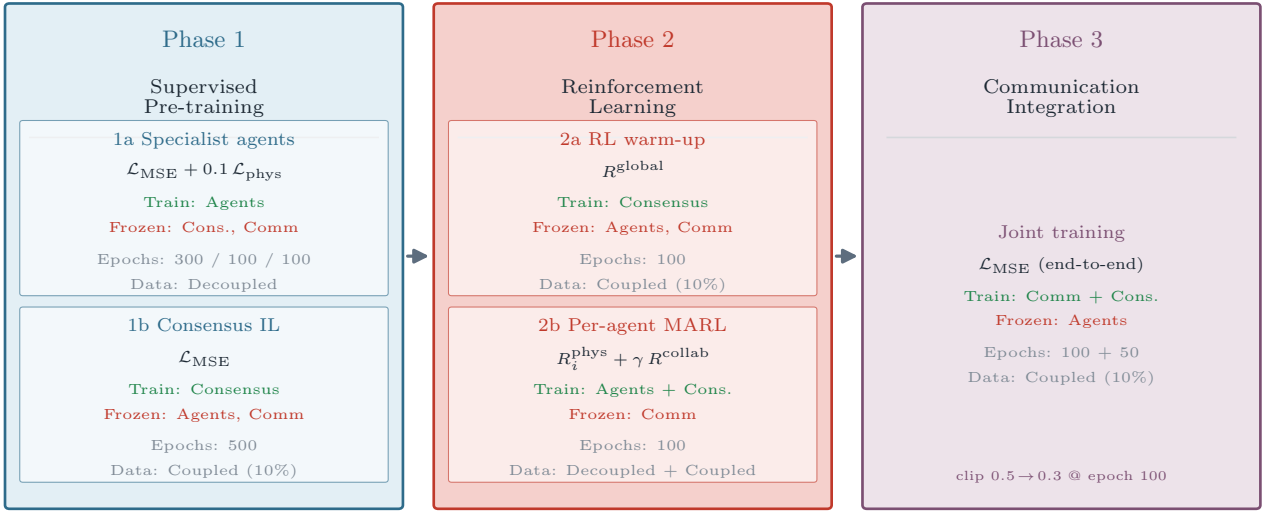
Phase 1
Supervised Pre-training
1a Specialist agents
$\mathcal{L}_{\text{MSE}} + 0.1\,\mathcal{L}_{\text{phys}}$
Train: Agents
Frozen: Cons., Comm
Epochs: 300 / 100 / 100
Data: Decoupled
1b Consensus IL
$\mathcal{L}_{\text{MSE}}$
Train: Consensus
Frozen: Agents, Comm
Epochs: 500
Data: Coupled (10%)
Phase 2
Reinforcement Learning
2a RL warm-up
R^{global}
Train: Consensus
Frozen: Agents, Comm
Epochs: 100
Data: Coupled (10%)
2b Per-agent MARL
$R_i^{\text{phys}} + \gamma\, R^{\text{collab}}$
Train: Agents + Cons.
Frozen: Comm
Epochs: 100
Data: Decoupled + Coupled
Phase 3
Communication Integration
Joint training
$\mathcal{L}_{\text{MSE}}$ (end-to-end)
Train: Comm + Cons.
Frozen: Agents
Epochs: 100 + 50
Data: Coupled (10%)
clip $0.5 \rightarrow 0.3$ @ epoch 100